

# Multi-Modal Drone Classification Using Visual Trajectories and Deep Feature Fusion

Yashwanth Hudumula

Department of Computer Science and Engineering,  
Blekinge Institute of Technology, Karlskrona, Sweden  
Email: yahu25@student.bth.se

**Abstract**—Reliable classification of aerial objects — birds, drones, and irrelevant background objects — is a critical task for safety-critical applications including airspace monitoring and counter-unmanned aerial vehicle (UAV) systems. This paper presents a multimodal deep learning approach that jointly exploits visual appearance from image sequences and kinematic information from 2D and 3D trajectory data for this three-class classification problem. Our proposed model, *img\_traject\_combined*, combines a pretrained EfficientNet-B0 image sequence encoder with a Transformer-based trajectory sequence encoder and a hand-crafted kinematic feature MLP, fused through a learned gated multimodal fusion mechanism. On the Visgrid classification dataset, our model achieves a macro F1-score of 0.9392 on the validation set and 0.8919 on the held-out test set, with a macro AUC of 0.9789. We compare our approach against three baselines — image-only, trajectory-only, and simple concatenation fusion — and conduct four ablation studies to measure the contribution of each component. Our results show that gated multimodal fusion consistently outperforms unimodal and naively fused alternatives, and that FFT-based frequency features and 3D trajectory data both improve test-set generalization.

## I. INTRODUCTION AND WORKLOAD DISTRIBUTION

With the growing usage of UAVs, there exists an immediate need to develop effective detection and classification methods that can distinguish between UAVs and natural aerial entities such as birds. This problem is harder than standard object recognition tasks because small flying objects at long range are difficult to resolve visually, cluttered backgrounds can occlude or camouflage them, and heterogeneous data modalities must be processed simultaneously.

Recent progress in deep learning technology has been very successful in different perceptual applications [1]. Multimodal fusion has been recently explored, revealing that processing visual information alongside motion information provides more accurate results compared to unimodal approaches [2]. This work investigates that claim by applying it to the task of aerial object classification on the Visgrid classification dataset [20].

The main contributions of our report are:

- A multimodal gated fusion model using sequences of images, trajectories, and hand-crafted kinematic features for the three-class aerial object classification problem.
- Systematic evaluation against three baselines: image-only, trajectory-only, and naive concatenation fusion.
- Four ablation studies demonstrating the significance of FFT-based frequency features, 3D trajectory data, pre-training, and decision threshold calibration.

- Experimental results showing that gated multimodal fusion outperforms unimodal and naively fused models on an imbalanced aerial dataset.

## II. PROBLEM DEFINITION

### A. Task Formulation

For the Visgrid classification dataset, the objective is to classify tracked aerial objects into one of three categories: BIRD, DRONE, or IRRELEVANT. For each tracked object, two types of information are provided: (i) thumbnails captured by one or more cameras (for image-based detection), and (ii) 2D trajectories from each camera as well as 3D trajectories generated by triangulation across multiple cameras.

The primary evaluation metric is macro-averaged F1-score, which assigns equal weight to each class regardless of its frequency in the dataset. In this challenge, failing to detect a drone is considered particularly costly.

### B. Challenges

The dataset presents several challenges:

**Low-resolution imagery.** All thumbnails are  $32 \times 32$  pixels. Distinguishing visually similar objects is very difficult at this resolution; for example, a bird with outstretched wings can closely resemble a drone.

**Class imbalance.** The class distribution is uneven: IRRELEVANT objects appear far less frequently than birds and drones, which typically degrades classifier performance on minority classes.

**Insufficient dataset size.** With only approximately 572 labeled samples for training and validation, training a large neural network from scratch is not feasible.

**Variable sequence length.** Image sequences and trajectory data vary in length across samples, requiring careful padding to avoid introducing noise into the model.

## III. BACKGROUND AND RELATED WORK

### A. Convolutional Neural Networks for Image Classification

Since AlexNet's groundbreaking results [3], CNNs have been the dominant architecture for image classification. Subsequent models such as VGGNet [4], ResNet [5], and EfficientNet [6] advanced the accuracy–efficiency trade-off further. EfficientNet-B0, the backbone used in our image sequence encoder, offers a strong balance through its compound scaling

strategy that jointly increases network depth, width, and input resolution.

ImageNet-pretrained weights can substantially boost performance when fine-tuning on small target datasets [7], which motivated our choice of ImageNet initialization and partial freezing of the early layers of EfficientNet-B0.

### B. Transformers for Sequence Modelling

Originally designed for NLP, the Transformer architecture [8] has since been applied to other sequential data types, including time series classification [9] and trajectory prediction [10]. Self-attention allows the model to capture long-range dependencies within a sequence, which is important for analysing temporal patterns such as the flapping frequency of birds or the hovering behaviour of drones.

### C. Multimodal Fusion

Multimodal learning integrates information from multiple input modalities [2]. Early, late, and intermediate fusion strategies have all been used successfully to combine signals from different sensors. Attention-based and gated fusion models [11] have demonstrated advantages over naive concatenation by learning to weight each modality according to its informativeness on a per-instance basis.

Several prior works have addressed UAV and aerial object classification, including radar-based [18] and video-based [15] approaches. More recently, camera and motion sensor fusion has been explored [17], though multimodal datasets for this task remain scarce.

### D. Drone vs. Bird Classification

Differentiating drones from birds has attracted growing interest due to airspace safety concerns. Svanström et al. [15] proposed real-time drone classification from video, and Coluccia et al. [16] surveyed multi-sensor detection approaches. Motion-based cues, particularly periodic flapping and directional consistency, have been identified as discriminative complements to visual appearance features.

## IV. METHODOLOGY

### A. Overview

Our proposed model, *img\_traject\_combined*, processes three parallel input streams: image sequences, trajectory sequences, and a vector of kinematic features. Each stream is encoded by a dedicated neural module into a fixed-dimensional embedding of size  $d = 128$ . These three embeddings are then combined by a learned gated multimodal fusion module that produces a per-instance weighted sum, which is passed to a classification head. Figure 1 illustrates the overall architecture.

### B. Image Sequence Encoder

The raw thumbnail images ( $32 \times 32$ ) are scaled up to ( $96 \times 96$ ) by applying bilinear interpolation before feeding them into the backbone. For feature extraction, we employ EfficientNet-B0 [6], pretrained on ImageNet [3]. The first

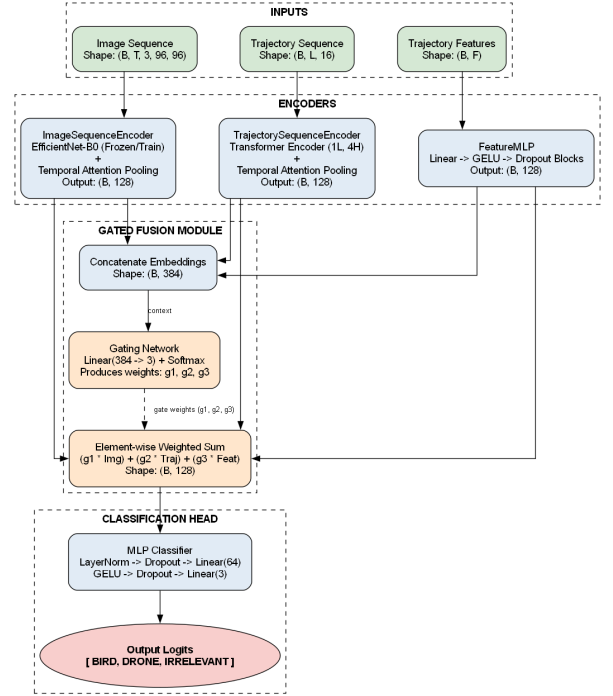


Fig. 1: Proposed multimodal gated fusion architecture.

four of the nine EfficientNet-B0 layers are kept frozen during training, preserving low-level features learned from ImageNet, while the last five layers are fine-tuned with a reduced learning rate.

A maximum of  $T = 16$  frames per object is randomly sampled during training; uniform sampling is used during evaluation. Each frame is embedded into a 1280-dimensional feature vector via EfficientNet's global average pooling layer and then projected to  $d = 128$  dimensions with a GELU-activated linear layer and dropout. Temporal attention pooling over these frame embeddings yields:

$$\mathbf{e}_{\text{img}} = \sum_{t=1}^T \alpha_t \mathbf{h}_t, \quad \alpha_t = \frac{\exp(w^\top \mathbf{h}_t)}{\sum_{t'} \exp(w^\top \mathbf{h}_{t'})} \quad (1)$$

where  $\mathbf{h}_t$  is the projected frame embedding at time  $t$ ,  $w$  is a learnable weight vector, and padding positions are masked out before softmax normalization.

### C. Trajectory Sequence Encoder

The trajectory sequence is formed by concatenating the 2D angular trajectory and the 3D trajectory data into a 5-dimensional position vector per time step. Velocity, acceleration, and instantaneous speed are appended via finite differencing, yielding a 16-dimensional feature vector per time step.

Each sequence is padded or truncated to a fixed length of  $L = 200$ . Trajectory sequences are encoded by a single-layer Transformer encoder [8] with embedding dimension  $d = 128$ , four attention heads, and pre-layer normalization. Positional encoding is applied before the Transformer. Attention pooling as per Equation (1) is used to extract the fixed-dimensional embedding  $\mathbf{e}_{\text{traj}}$ .

#### D. Kinematic Feature Encoder

In parallel with the trajectory sequence encoder, we compute a vector of hand-crafted kinematic features designed to capture motion statistics that may be difficult for the Transformer to learn from short sequences. These include:

- Statistical moments (mean, standard deviation) of position, velocity, acceleration, and jerk over the full trajectory.
- Speed statistics including mean, standard deviation, maximum, and 25th/75th percentiles (capturing the higher speed variability of birds).
- Curvature statistics (mean, standard deviation, maximum direction change rate).
- Vertical oscillation frequency and amplitude (birds exhibit periodic vertical motion due to flapping).
- FFT-based frequency features computed on the speed signal: dominant frequency bin, maximum magnitude, mean magnitude, and standard deviation (capturing the periodic flapping frequency of birds).
- Path geometry features: total path length, displacement, and straightness ratio (drones tend to follow straighter paths than birds).
- Multi-view consistency features: standard deviation of the above features across cameras.

These features are processed by a three-layer MLP with GELU activations, dropout, and batch normalization on the input to handle feature scale differences. The final layer projects to  $d = 128$ , producing embedding  $\mathbf{e}_{\text{feat}}$ .

#### E. Gated Multimodal Fusion

The three embeddings  $\mathbf{e}_{\text{img}}$ ,  $\mathbf{e}_{\text{traj}}$ ,  $\mathbf{e}_{\text{feat}}$  are concatenated and passed through a two-layer network with softmax output to produce per-instance modality gates  $\mathbf{g} \in R^3$ :

$$\mathbf{g} = \text{softmax}(W_g[\mathbf{e}_{\text{img}}; \mathbf{e}_{\text{traj}}; \mathbf{e}_{\text{feat}}] + b_g) \quad (2)$$

$$\mathbf{e}_{\text{fused}} = g_1 \mathbf{e}_{\text{img}} + g_2 \mathbf{e}_{\text{traj}} + g_3 \mathbf{e}_{\text{feat}} \quad (3)$$

This mechanism allows the network to adaptively rely on different modalities depending on the available information for each object — for example, up-weighting trajectory information for objects with long, distinctive motion patterns, or visual features for objects with distinctive appearance.

#### F. Classification Head and Loss Function

The fused embedding is passed through a two-layer classification head with LayerNorm, GELU activation, and dropout, producing logits over the three classes.

We use Focal Loss [12] with label smoothing ( $\epsilon = 0.1$ ) to address class imbalance. Class weights are set to  $[1.0, 2.5, 2.0]$  for BIRD, DRONE, and IRRELEVANT respectively, reflecting both the drone detection priority and the class distribution. The focal weighting parameter is set to  $\gamma = 1.5$ .

TABLE I: Dataset Statistics

| Split      | Total | BIRD     | DRONE    | IRRELEVANT |
|------------|-------|----------|----------|------------|
| Train      | ~327  | majority | majority | minority   |
| Validation | 121   | 53       | 39       | 29         |
| Test       | 124   | 55       | 40       | 29         |
| Val+Test   | 245   | 108      | 79       | 58         |

#### G. Decision Threshold Calibration

After training, we apply post-hoc decision threshold calibration via a grid search over per-class probability bias terms to maximise macro F1-score on the validation set. This corrects the tendency of neural classifiers to underestimate confidence for minority classes:

$$\hat{y} = \arg \max_k [\text{softmax}(\mathbf{z})_k \cdot \exp(b_k)] \quad (4)$$

where  $b_k$  are the calibration biases found by grid search, with  $b_{\text{BIRD}} = 0$  fixed and  $b_{\text{DRONE}}$ ,  $b_{\text{IRRELEVANT}}$  searched over  $[-0.5, 1.5]$  and  $[-0.5, 0.5]$  respectively.

#### H. Training Procedure

All models are trained using AdamW with weight decay of  $10^{-3}$  and gradient clipping set to 2.0. The backbone uses a learning rate of  $10^{-5}$ , while the remaining layers use  $10^{-4}$ . A cosine annealing scheduler with warm restarts ( $T_0 = 20$ ) is applied during training. Early stopping is triggered if the validation macro F1-score does not improve for 15 epochs, and the best-performing checkpoint is saved. Training augmentations include random cropping, horizontal flipping, colour jitter, grayscale conversion, and random erasing.

### V. EXPERIMENTAL SETUP

#### A. Dataset

The Visgrid classification dataset consists of tracked aerial objects represented by sequences of  $32 \times 32$  images collected by one or multiple cameras. The classes are BIRD, DRONE, and IRRELEVANT. The dataset was provided by the Visgrid company as part of the Naval Defense Tech Hackathon held in Karlskrona. It is divided into training, validation, and test splits. The validation set contains 121 objects (53 birds, 39 drones, 29 irrelevant), and the test set contains 124 objects (55 birds, 40 drones, 29 irrelevant).

Table I summarises the dataset statistics.

#### B. Baseline Models

The proposed model is benchmarked against three baselines of increasing fusion complexity:

**Image-only (img\_only\_baseline):** Uses only the EfficientNet-B0 image sequence encoder with temporal attention pooling. No trajectory data is used.

**Trajectory-only (traj\_only\_baseline):** Uses only the Transformer-based trajectory sequence encoder and the kinematic feature MLP. No image information is used.

TABLE II: Test Set Results Baseline Comparison

| Model               | Macro F1      | BIRD | DRONE | IRREL | AUC           |
|---------------------|---------------|------|-------|-------|---------------|
| Gated Fusion (ours) | <b>0.8919</b> | 0.89 | 0.85  | 0.93  | <b>0.9789</b> |
| Image-Only          | 0.8909        | 0.87 | 0.88  | 0.92  | 0.9784        |
| Trajectory-Only     | 0.8883        | 0.91 | 0.85  | 0.91  | 0.9530        |
| Concat Fusion       | 0.8263        | 0.79 | 0.77  | 0.92  | 0.9167        |

**Concatenation-based fusion:** Uses the same three encoders as the proposed model, but replaces the gated fusion module with direct concatenation of the embeddings before the classification head.

### C. Ablation Studies

To measure the contribution of each individual component, we conduct four ablation studies:

**Ablation A (Without FFT features):** All FFT-based frequency features are removed from the kinematic feature set to assess their discriminative value.

**Ablation B (Without pretrained backbone):** EfficientNet-B0 is initialised randomly rather than from ImageNet weights, to quantify the benefit of transfer learning under limited data.

**Ablation C (Without decision threshold calibration):** Predictions are made by taking the argmax of the raw logits, without post-hoc calibration, to evaluate the impact of this post-processing step.

**Ablation D (Without 3D trajectory):** Only 2D trajectory data is used, omitting the 3D component. This ablation also simulates a degraded scenario where multiple cameras fail.

### D. Metrics for Evaluation

All experiments report macro F1-score (primary metric), class-wise precision, recall, and F1-score, and macro AUC (one-vs-rest). Macro F1-score is used as the primary metric because it penalises poor performance on any individual class, including the minority class (IRRELEVANT).

### E. Implementation Details

Models are implemented in Python using the PyTorch library [14] and trained on a single GPU (NVIDIA GeForce series). The random seed is fixed at 42. EfficientNet-B0 weights are loaded from `torchvision.models.EfficientNet_B0_Weights.DEFAULT` (pretrained on ImageNet). Training runs for a maximum of 60 epochs with early stopping (patience 15). Batch size is 16 for multimodal models and 32 for the trajectory-only baseline.

## VI. RESULTS AND DISCUSSION

### A. Baseline Comparison

Table II reports test set results for all four models. Our proposed gated multimodal fusion model achieves the best macro F1-score of 0.8919, while the concatenation baseline scores considerably lower at 0.8263.

**Image-only model.** Despite operating on extremely low-resolution ( $32 \times 32$ ) thumbnails, the image-only baseline

achieves a macro F1-score of 0.8909, close to our full model. This highlights the strong contribution of the pretrained EfficientNet-B0 backbone. However, the absence of motion information limits its BIRD F1-score (0.87) relative to the trajectory-only model.

**Trajectory-only model.** Using motion alone yields a macro F1-score of 0.8883, comparable to the visual-only approach without any pixel information. This confirms that object kinematics are highly discriminative: drones tend to follow linear paths at uniform speeds, while birds exhibit speed variability and periodic vertical motion from wing flapping. The trajectory encoder achieves the highest BIRD F1-score of all models (0.91).

**Concatenation fusion.** The concatenation baseline performs the worst (0.8263), with notably low BIRD and DRONE F1-scores of 0.79 and 0.77 respectively. This underperforms both unimodal baselines, consistent with the known limitations of late concatenation fusion, which lacks a mechanism to adaptively weight modalities.

**Proposed gated multimodal fusion.** Our approach achieves the best macro F1-score across all models. The gated fusion mechanism allows the model to weight modalities adaptively: objects with distinctive visual appearance rely more on the image encoder, while those with long, characteristic trajectories rely more on the trajectory encoder. Our model also achieves the highest IRRELEVANT F1-score (0.93) and strong DRONE recall (0.95).

### B. Comparison with Published Methods

Table III contextualises our results against published aerial object classification methods. Because these methods use different datasets, sensor modalities, and task definitions, direct numerical comparison is not possible; the table is intended to situate our work qualitatively within the field.

TABLE III: Comparison with Published Aerial Object Classification Methods

| Method                  | Modality                 | Task            | Best Result       |
|-------------------------|--------------------------|-----------------|-------------------|
| Svanström et al. [15]   | et Video + thermal audio | Drone detection | F1: 0.7601        |
| Coluccia et al. [16]    | et Video (DL)            | Drone bird      | Avg. P: 0.73      |
| Hammer et al. [17]      | et Radar $\mu$ -Doppler  | UAV bird        | Qualitative       |
| Rahman & Robertson [18] | Radar K/W-band           | UAV bird        | Sig. analysis     |
| <b>Ours</b>             | <b>Image + traj.</b>     | <b>3-class</b>  | <b>F1: 0.8919</b> |

Our macro F1-score of 0.8919 is competitive with or exceeds the published figures above, despite operating on extremely low-resolution ( $32 \times 32$ ) thumbnails with only  $\sim 327$  training samples. Note that Svanström et al. address a detection task (localisation plus classification) rather than pure classification, making their F1-score less directly comparable.

Hammer et al. and Rahman & Robertson characterise radar signatures rather than reporting a single classification metric, confirming that camera-and-trajectory fusion is a complementary and effective modality for this problem.

### C. Ablation Study Results

Table IV reports ablation results evaluated on both validation and test sets.

TABLE IV: Ablation Study Results

| Ablation                             | Val F1 | $\Delta$ val | Test F1 | Test AUC |
|--------------------------------------|--------|--------------|---------|----------|
| Full model (baseline)                | 0.9392 | –            | 0.8919  | 0.9789   |
| B: No pretrained backbone            | 0.9614 | +0.022       | 0.8893  | 0.9638   |
| C: No decision threshold calibration | 0.9390 | –0.0002      | 0.8851  | 0.9789   |
| A: No FFT features                   | 0.9392 | 0.000        | 0.8847  | 0.9754   |
| D: No 3D trajectory                  | 0.9410 | +0.002       | 0.8707  | 0.9683   |

**Ablation A (No FFT features).** Removing FFT-based frequency features leaves the validation macro F1-score unchanged (0.9392), but reduces the test macro F1-score from 0.8919 to 0.8847 and the macro AUC from 0.9789 to 0.9754. This indicates that FFT features carry discriminative information that generalises beyond the validation set.

**Ablation B (No pretrained backbone).** The scratch-trained model attains a higher validation macro F1-score (0.9614 vs. 0.9392), yet nearly identical test scores (0.8893 vs. 0.8919). The higher validation performance reflects overfitting to the small validation set rather than genuine generalisation. The test scores differ by only 0.0026 points.

**Ablation C (No decision threshold calibration).** Calibration yields a marginal validation improvement of 0.0002 (0.9390  $\rightarrow$  0.9392) but a more meaningful test improvement from 0.8851 to 0.8919. Given that calibration requires no additional training time, this gain is worthwhile.

**Ablation D (No 3D trajectory).** Excluding 3D trajectory data slightly improves the validation macro F1-score (+0.0018), but causes the largest test-set drop of all ablations, from 0.8919 to 0.8707. Although 2D angular trajectories are sufficient to fit the small validation set, 3D trajectory data provides richer representations of object dynamics — encoding height, angle, and distance — that are more discriminative for distinguishing drone and bird flight patterns.

### D. Validation vs. Test Set Performance

All models show a gap of 0.04–0.08 between validation and test macro F1-scores. Given the small size of each split (121 and 124 samples respectively) and the stochastic nature of the data split, this discrepancy is expected and does not indicate overfitting.

The DRONE category is the most stable: the full model achieves a test recall of 0.95, confirming reliable drone detection. This is the most safety-critical result in this application context.

### E. Failure Analysis

Analysis of the validation confusion matrix reveals two main error types. First, some BIRD instances are misclassified

as DRONE — these correspond to birds flying in a straight line at high speed, mimicking the controlled flight path of a drone. Second, some DRONE instances are misclassified as BIRD — these are drones exhibiting erratic, low-altitude flight resembling bird behaviour.

The IRRELEVANT category is consistently well classified (F1-score 0.93–0.98 across all models), suggesting that irrelevant objects possess sufficiently distinctive visual or kinematic features to be reliably separated from aerial targets.

### F. Overfitting Analysis

Training loss curves show stable convergence in all cases. The gap between training and validation loss stabilises around epoch 20–30, confirming effective regularisation. Cosine annealing with warm restarts helps avoid local minima, and the combination of label smoothing, focal loss, dropout ( $p = 0.3$ – $0.4$ ), and weight decay ( $10^{-3}$ ) prevents overfitting on the small training set.

Early stopping triggers between epoch 10 and epoch 40; the full model stops at epoch 35 on average. A patience of 15 epochs was found appropriate for this dataset size.

## VII. CONCLUSION AND FUTURE WORK

This paper introduced a multimodal gated fusion system for aerial object classification that jointly processes image sequences and trajectory data to distinguish BIRD, DRONE, and IRRELEVANT objects. The system achieves a macro F1-score of 0.8919 and a macro AUC of 0.9789 on the test set, outperforming image-only, trajectory-only, and naive concatenation baselines. Ablation studies confirm the contributions of gated multimodal fusion, 3D trajectory data, FFT-based frequency features, and decision threshold calibration.

Several directions for future work remain. The trajectory sequence encoder could be extended with camera-aware encoding that processes each camera’s trajectory independently before cross-camera attention. The uniform frame sampling strategy could be replaced with motion-aware sampling guided by optical flow magnitude to select the most informative frames. An asymmetric cost-sensitive loss could further increase DRONE recall. Finally, trajectory augmentation methods such as jittering, speed scaling, and rotation could improve generalisation on limited data.

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## APPENDIX

## SUPPLEMENTARY MATERIAL

This appendix provides complete confusion matrices, per-class metric charts, and training curves for all models and ablation variants reported in the main paper.

### A. Normalized Confusion Matrices – Proposed Model and Baselines

Figures 2–9 show row-normalized confusion matrices on the validation and test sets for all four models. Row normalization expresses each entry as the fraction of true-class samples assigned to each predicted class, making per-class recall directly readable from the diagonal.

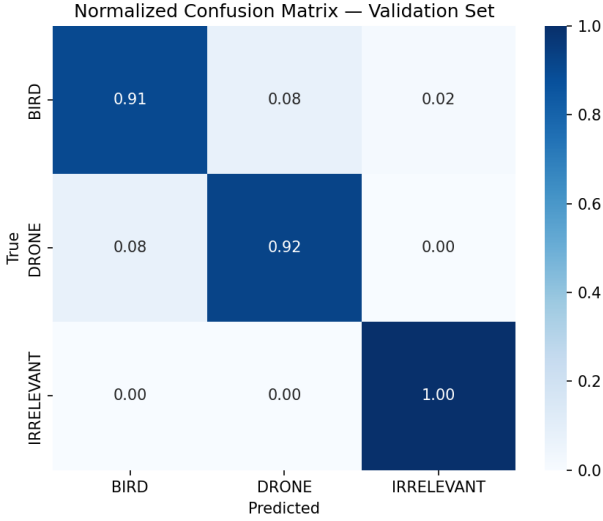


Fig. 2: Validation CM —Gated Fusion (proposed).

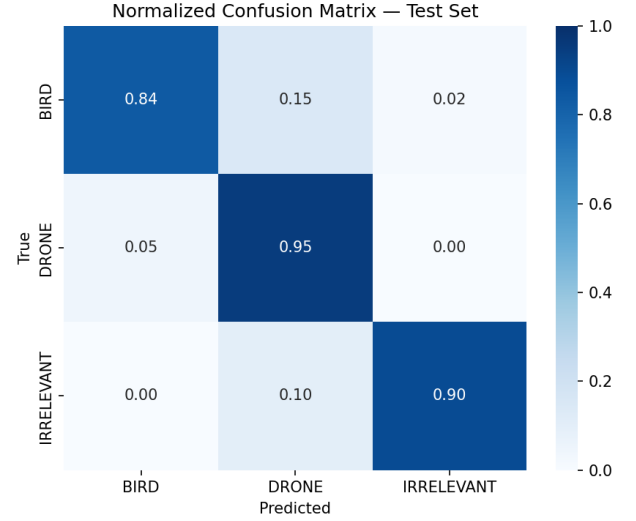


Fig. 3: Test CM Gated Fusion (proposed).

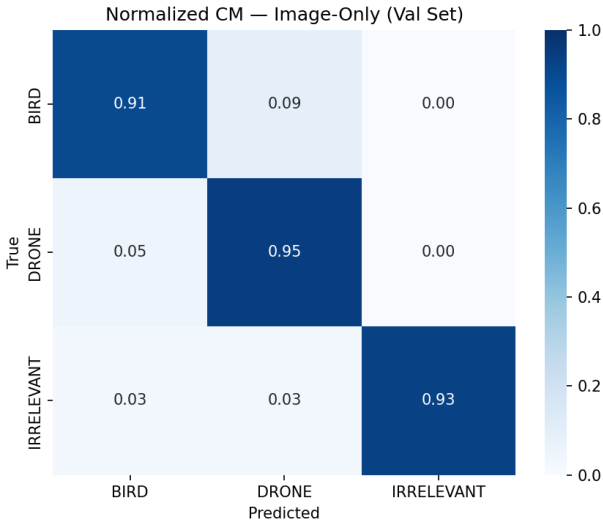


Fig. 4: Validation CM Image-Only baseline.

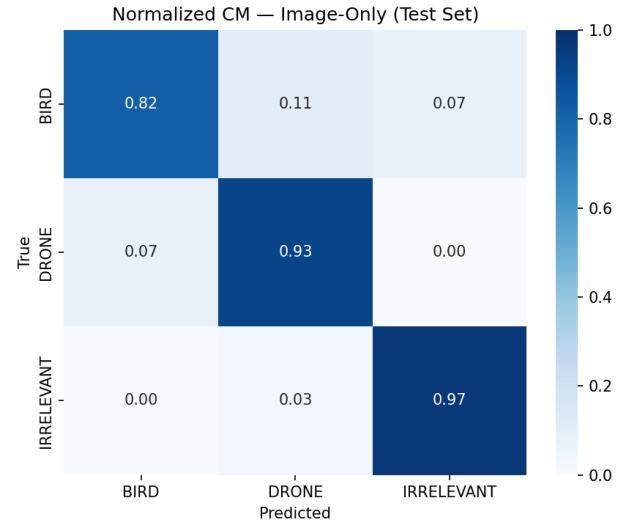


Fig. 5: Test CM Image-Only baseline.

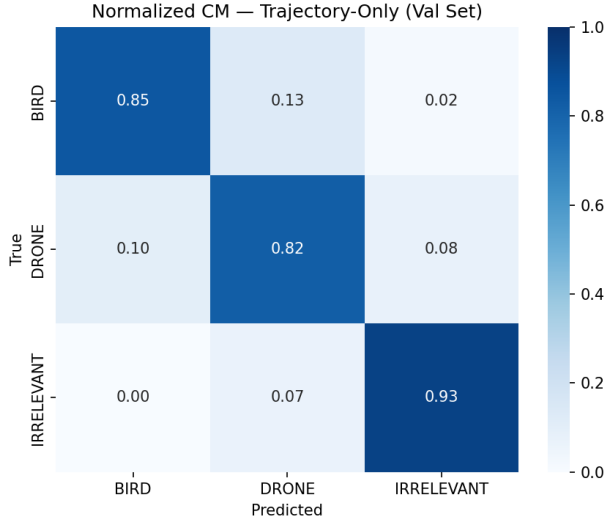


Fig. 6: Validation CM Trajectory-Only baseline.

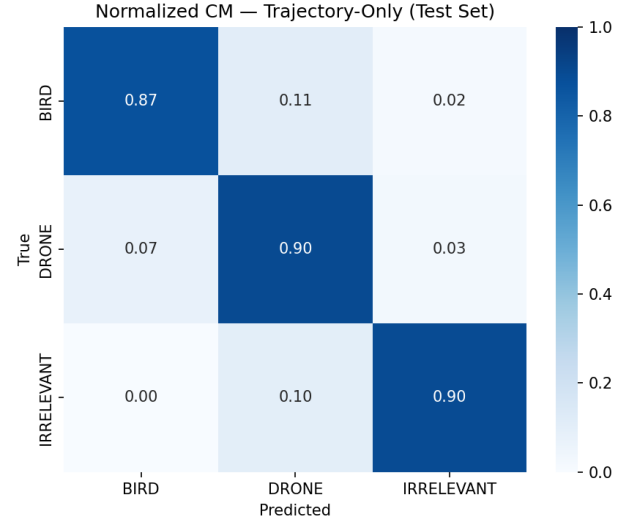


Fig. 7: Test CM Trajectory-Only baseline.

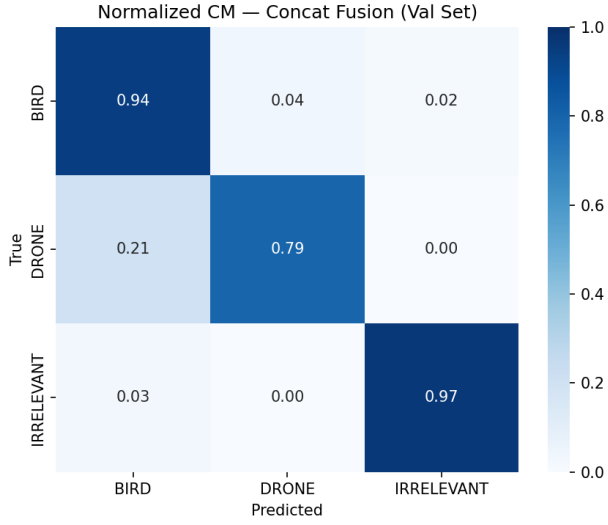


Fig. 8: Validation CM Concatenation Fusion baseline.

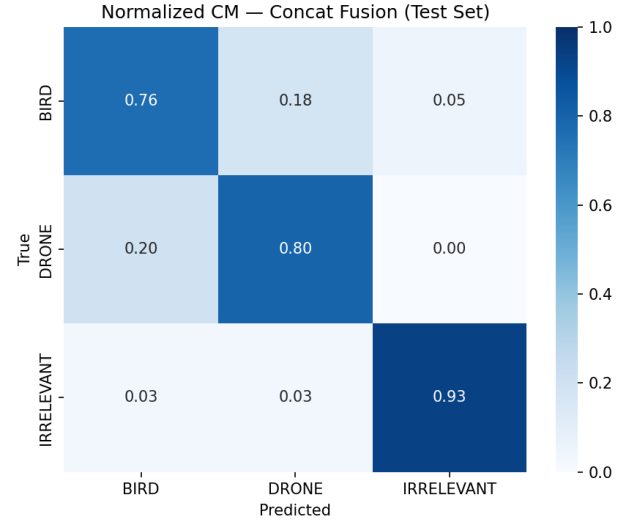


Fig. 9: Test CM Concatenation Fusion baseline.

### B. Normalized Confusion Matrices Ablation Studies

Figures 10–17 show confusion matrices for each ablation variant (A–D) on both splits. Comparing these to Figures 2–3 isolates the effect of each removed component on the per-class error distribution.



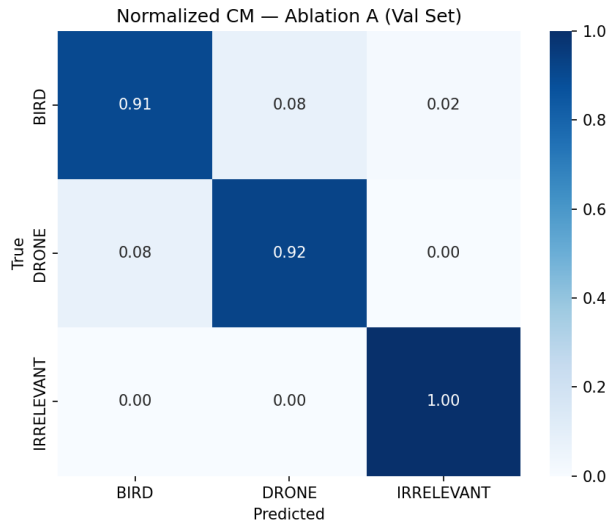


Fig. 10: Validation CM Ablation A (No FFT features).

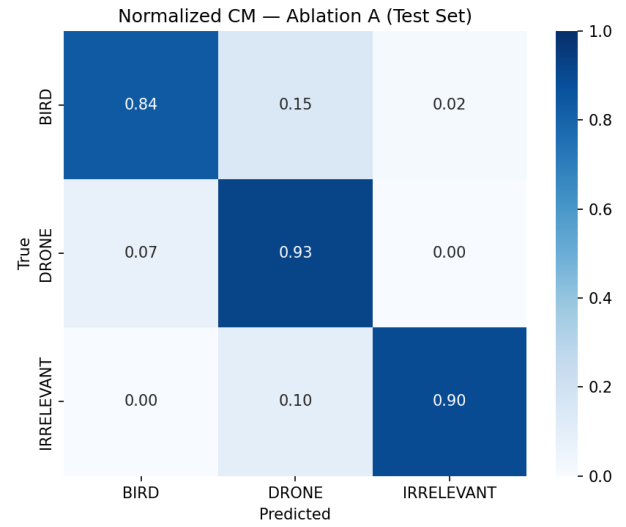


Fig. 11: Test CM Ablation A (No FFT features).

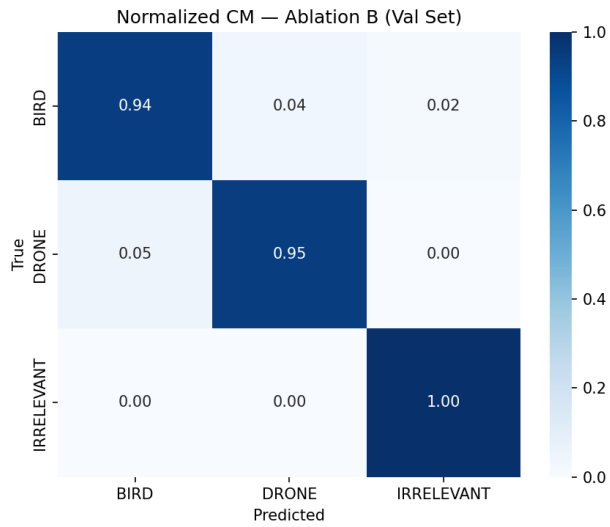


Fig. 12: Validation CM Ablation B (No pretrained backbone).

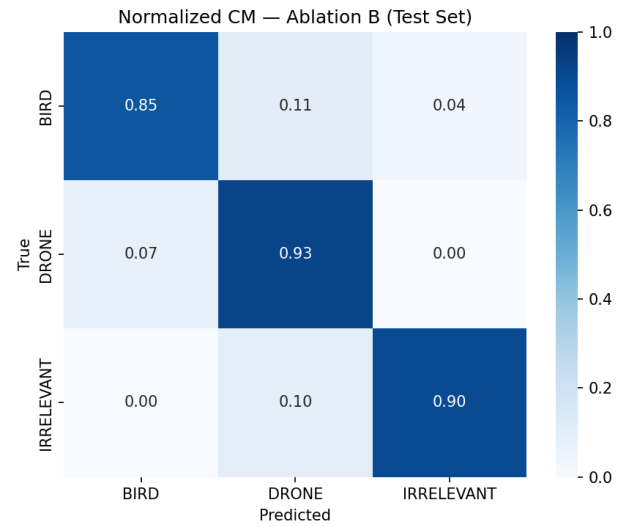


Fig. 13: Test CM Ablation B (No pretrained backbone).

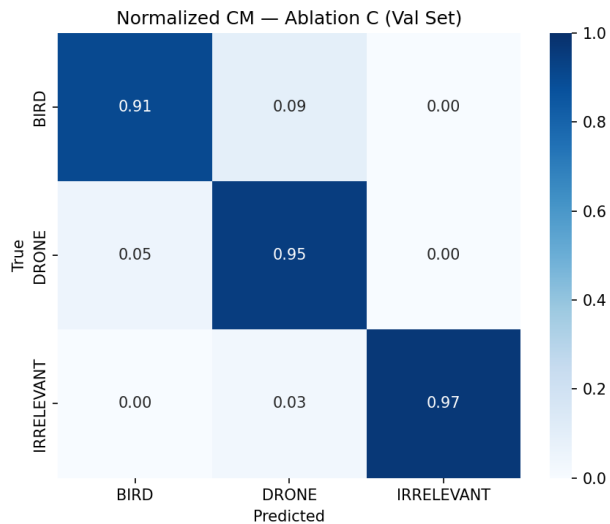


Fig. 14: Validation CM Ablation C (No decision threshold calibration).

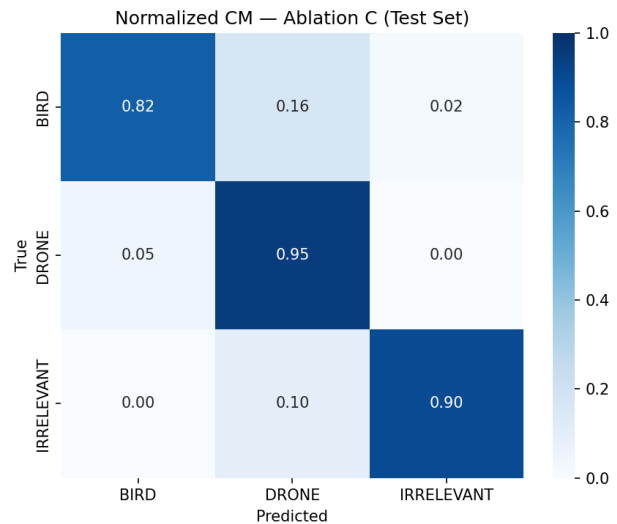


Fig. 15: Test CM Ablation C (No decision threshold calibration).

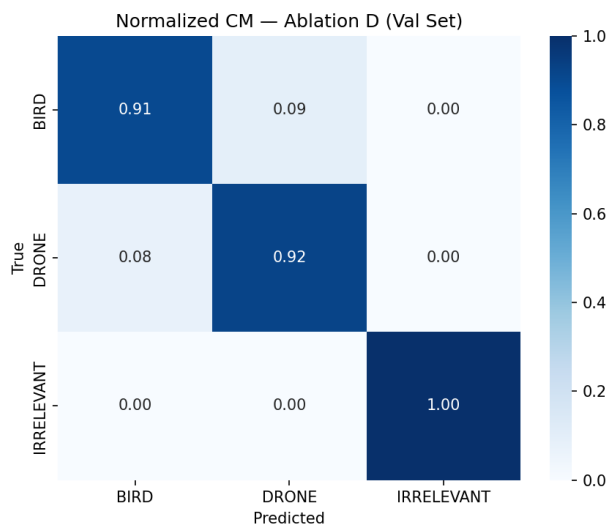


Fig. 16: Validation CM Ablation D (No 3D trajectory).

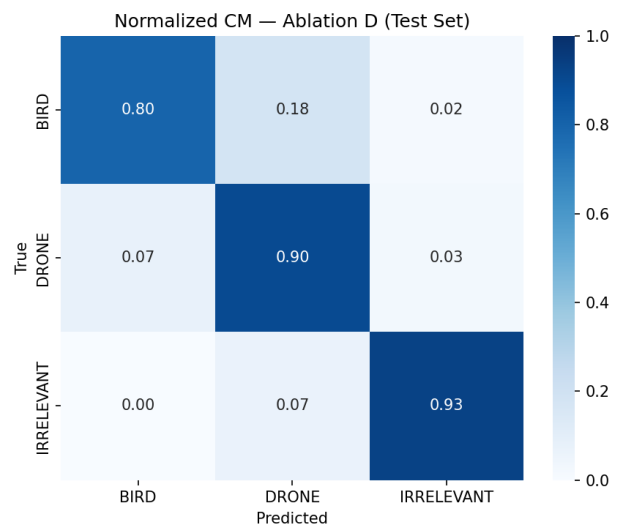


Fig. 17: Test CM Ablation D (No 3D trajectory).

### C. Per-Class Precision, Recall, and F1 Baselines

Figures 18–21 break down precision, recall, and F1-score per class (BIRD, DRONE, IRRELEVANT) for each model. These charts complement Table II in the main paper by making the per-class trade-offs between precision and recall visually explicit.

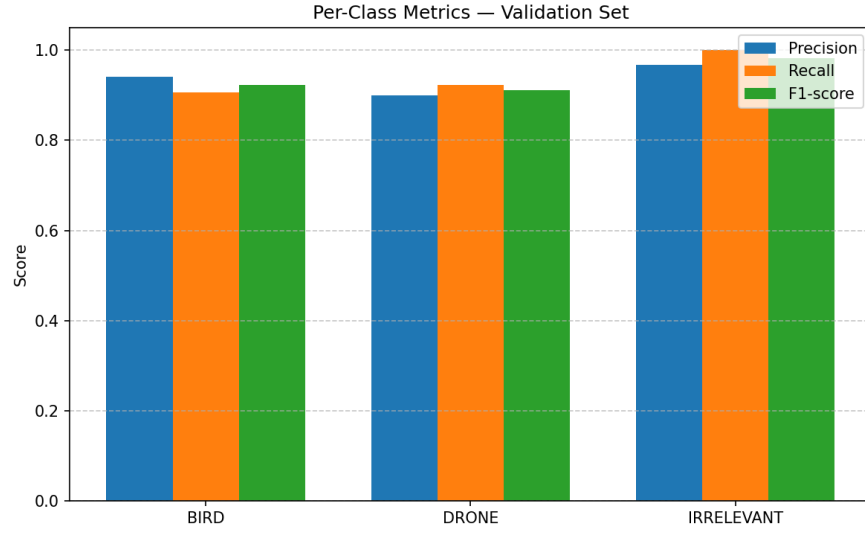


Fig. 18: Per-class precision, recall, and F1-score — Gated Fusion (proposed).

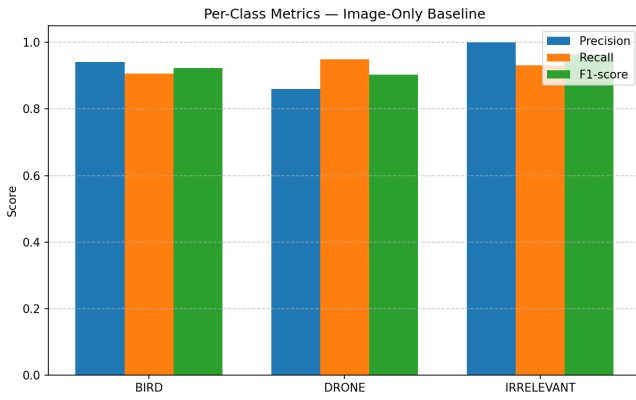


Fig. 19: Per-class metrics Image-Only baseline.

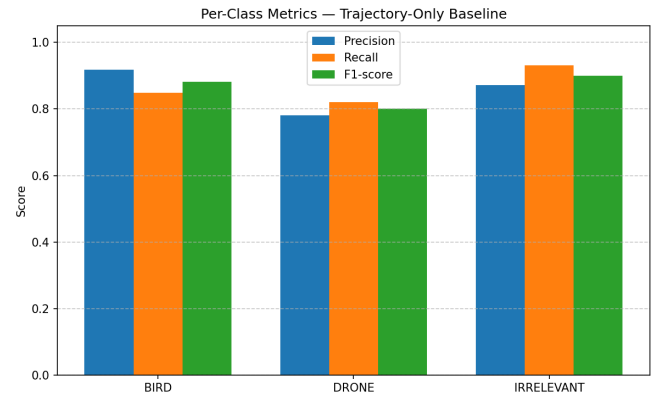


Fig. 20: Per-class metrics Trajectory-Only baseline.

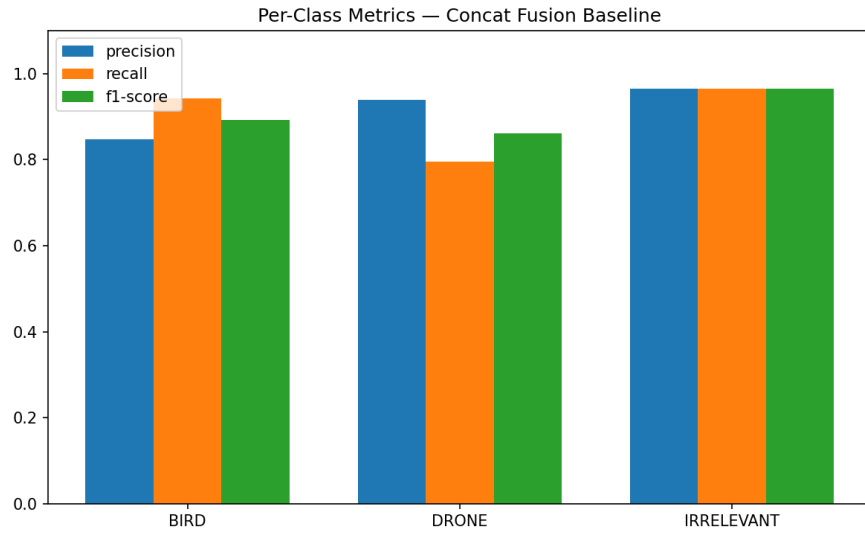


Fig. 21: Per-class metrics Concatenation Fusion baseline.

#### D. Training Curves Proposed Model and Baselines

Figures 22–25 show epoch-wise training loss, validation loss, and validation macro F1-score for all models. The vertical dashed line in each plot marks the early-stopping checkpoint selected for evaluation.

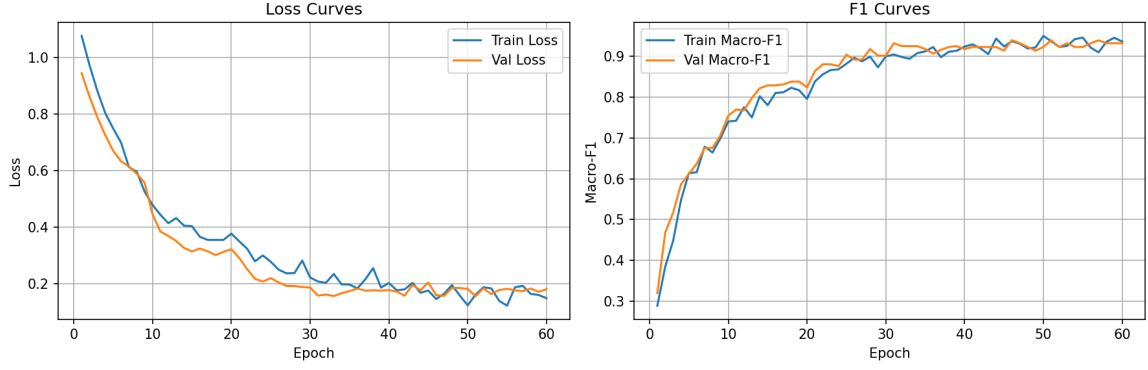


Fig. 22: Training curves Gated Fusion (proposed).

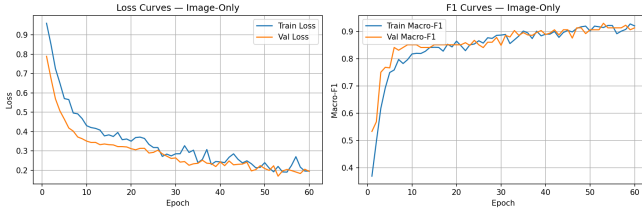


Fig. 23: Training curves Image-Only baseline.

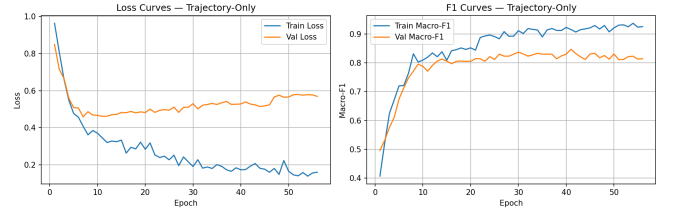


Fig. 24: Training curves Trajectory-Only baseline.

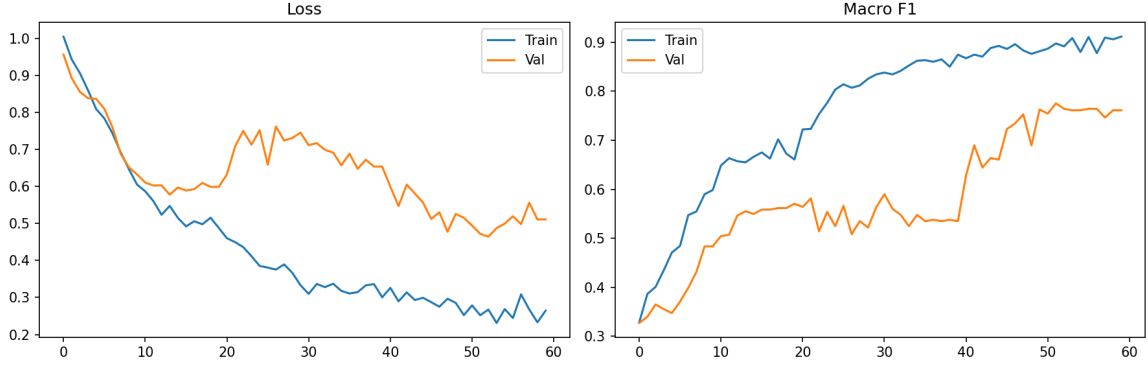


Fig. 25: Training curves Concatenation Fusion baseline.

### E. Training Curves Ablation Studies

Figures 26–28 show training dynamics for ablation variants A, B, and D. Ablation C (no decision threshold calibration) shares identical training curves with the full model since calibration is a post-hoc step.

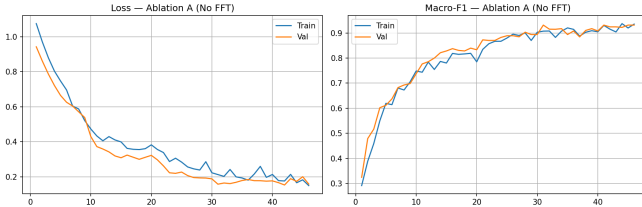


Fig. 26: Training curves Ablation A (No FFT features).

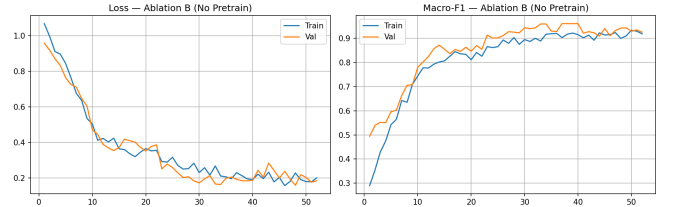


Fig. 27: Training curves Ablation B (No pretrained backbone). Note the higher validation macro F1-score at the higher validation set discussed in Section VI-B.

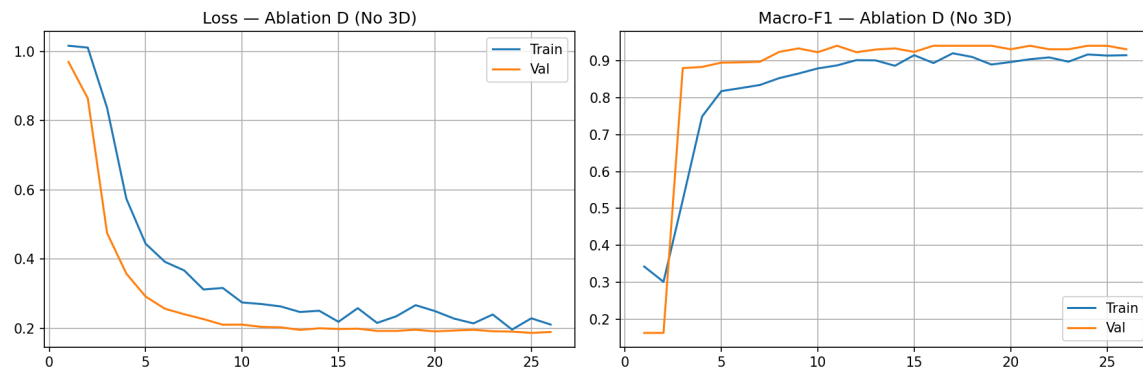


Fig. 28: Training curves Ablation D (No 3D trajectory). The faster convergence relative to the full model reflects the reduced input dimensionality of the trajectory sequence encoder.